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Studierichting: Informatica
Oefeningenreeks 4A nr. 70.4

$$\begin{aligned}\int x^n \ln(x) dx &= \frac{1}{n+1} \int \ln(x) d(x^{n+1}) = \frac{x^{n+1}}{n+1} \ln(x) - \frac{1}{n+1} \int x^{n+1} d(\ln(x)) \\ &= \frac{x^{n+1}}{n+1} \ln(x) - \frac{1}{n+1} \int x^n dx \\ &= \frac{x^{n+1}}{n+1} \ln(x) - \frac{x^{n+1}}{(n+1)^2} + c\end{aligned}$$

References